

CSE 3203
Computer Network

The OSI (Open Systems Interconnection) model is a universal conceptual framework created by the International Organization for Standardization (ISO). It standardizes telecommunication and computing functions into seven abstract layers, allowing diverse hardware and software to communicate efficiently and troubleshooting to be isolated.

The model divides network communication into seven distinct layers, which data travels down on the sending device and up on the receiving device:

7. Application Layer

The interface between the network and your software. It allows applications (e.g., web browsers, email clients) to access network services.

- Key Protocols: HTTP, FTP, SMTP, DNS

6. Presentation Layer

Ensures data is in a readable, usable format for the Application layer. It handles data translation, encryption, decryption, and compression.

- Key formats: SSL/TLS, JPEG, MPEG

5. Session Layer

Opens, manages, and closes communication sessions between applications. It coordinates dialogue and connection maintenance between two devices.

- Key Protocols: RPC, NetBIOS

4. Transport Layer

Manages end-to-end communication between devices. It breaks data into segments, ensures reliable delivery, and controls flow and error recovery.

- Key Protocols: TCP, UDP

3. Network Layer

Handles logical addressing and routing. It determines the best physical path for data to travel across different networks. [[1](#)]

- Key Devices: Routers

- Key Protocols: IP (IPv4/IPv6), ICMP [[1](#), [2](#)]

2. Data Link Layer

Establishes and terminates a connection between two physically connected nodes on a network. It packages raw data into "frames" and handles physical hardware addressing (MAC addresses).

- Key Devices: Switches, Bridges
- Key Protocols: Ethernet, ARP

1. Physical Layer

Transmits raw bitstreams—the physical, unstructured stream of 0s and 1s—over a physical medium (copper cables, fiber optics, or radio waves).

- Key Devices: Hubs, Repeaters, Cables

Physical Layer in OSI Model

The Physical Layer is the lowest OSI layer responsible for transmitting raw data bits through physical network hardware.

- Handles physical and electrical data transmission
- Includes cables, connectors, plugs, and receivers
- Transfers raw bits between devices
- Defines hardware standards and signal types

Physical Layer in OSI Model

Functions of Physical Layer

- The Physical Layer is responsible for sending raw data as bits over a physical medium.
- It converts data into signals that can travel through wires, fiber optics, or wireless channels (encoding) and turns these signals back into data at the receiver (decoding).
- It ensures signals are transmitted correctly and uses techniques like modulation to prepare the data for transmission and demodulation to retrieve it at the other end.
- This layer also decides how data flows (one-way, two-way alternately, or simultaneously) through transmission modes and controls the speed and timing of data transmission to keep everything running smoothly.

Physical Topologies

Physical topologies describe the physical arrangement of devices and cables in a network. Below are the different topologies used in Computer Networks

Line Configuration

- Point-to-Point configuration: In Point-to-Point configuration, there is a line (link) that is fully dedicated to carrying the data between two devices.
- Multi-Point configuration: In a Multi-Point configuration, there is a line (link) through which multiple devices are connected.

Need of Physical Layer in Security

In security, Physical Layer is essential because many attacks can occur before any software is involved. Threats at this level target the actual hardware or transmission medium.

- **Cable Tapping:** Hackers can intercept data by directly connecting to network cables.
- **Physical Access:** If someone gets into server rooms or hardware areas without permission, they can steal or damage equipment.
- **Wireless Signal Interception:** Attackers can capture Wi-Fi signals from outside using special tools.
- **Hardware Manipulation:** Someone might tamper with physical devices like routers or USB ports to secretly install harmful software.

Applications

- It ensures devices can transmit and receive raw data over physical mediums.
- It provides universal standards for cables, connectors, and signaling, ensuring compatibility.
- **Support for Various Media:** Works with wired (e.g., Ethernet) and wireless (e.g., Wi-Fi) technologies.

Data Link Layer in OSI Model

The data link layer is the second layer from the bottom in the [OSI](#) (Open System Interconnection) network architecture model.

- Responsible for the node-to-node delivery of data within the same local network.
- Major role is to ensure error-free transmission of information.
- Also responsible for encoding, decoding, and organizing the outgoing and incoming data.
- Considered as the most complex layer of the OSI model as it hides all the underlying complexities of the hardware from the other above layers.

Sub-Layers of The Data Link Layer

The data link layer is further divided into two sub-layers, which are as follows:

1. **Logical Link Control (LLC):** This sublayer of the data link layer deals with multiplexing, the flow of data among applications and other services, and LLC is responsible for providing error messages and acknowledgments as well.
2. **Media Access Control (MAC):** MAC sublayer manages the device's interaction, responsible for addressing frames, and also controls physical media access. The data link layer receives the information in the form of packets from the Network layer, it divides packets into frames and sends those frames bit-by-bit to the underlying physical layer.

Functions of The Data-link Layer

- **Framing:** Divides the stream of bits from the Network Layer into manageable, structured data units called *frames*. It adds a header and trailer to identify the start/end of data.
- **Physical Addressing (MAC):** Inserts the sender's and receiver's MAC addresses (unique hardware addresses) into the frame header to ensure data reaches the correct device.
- **Error Control:** Detects and corrects transmission errors (e.g., bit corruption) using techniques like the *Cyclic Redundancy Check (CRC)*, ensuring corrupt frames are discarded or retransmitted.
- **Flow Control:** Prevents a fast sender from overwhelming a slow receiver by regulating the pace of data transmission. [[1](#), [2](#)]
- **Access Control:** Coordinates how multiple devices share the same physical communication medium (e.g., via protocols like CSMA/CD or CSMA/CA) to prevent collisions

Limitations of Data Link Layer

- **Limited Scope:** It operates only within a local network and cannot handle end-to-end communication across different networks.
- **Increased Overhead:** Adding headers, trailers, and redundant data (for error correction) increases the size of transmitted data.
- **Error Handling Dependency:** While it can detect and correct some errors, it relies on upper layers for handling more complex issues.

- **No Routing Capability:** The Data Link Layer cannot make routing decisions. It only ensures delivery within the same network segment.
- **Resource Usage:** Flow control and error correction mechanisms may consume extra processing power and memory

Applications of Data Link Layer

- **Local Area Networks (LANs):** Enables reliable communication between devices within a local network using protocols like Ethernet (IEEE 802.3).
- **Wireless Networks (Wi-Fi):** Manages communication between devices in wireless networks via protocols like IEEE 802.11 hence, handling media access and error control.
- **Switches and MAC Addressing:** Facilitates the operation of switches by using MAC addresses to forward data frames to the correct device within the network.
- **Point-to-Point Connections:** Used in protocols like PPP (Point-to-Point Protocol) to establish and manage direct communication between two nodes.

Go Back N - Sliding Window Protocol

Go-Back-N (GBN) is a sliding window–based Automatic Repeat Request (ARQ) protocol, in which the sender can transmit multiple frames (up to a fixed window size) without waiting for an acknowledgement for each frame.

- If a frame is lost or corrupted, the receiver discards that frame.
- All subsequent out-of-order frames are also discarded.
- The sender retransmits the erroneous frame and all following frames in the current window.
- Example: If frames 1 to 5 are sent and frame 3 is lost, the sender retransmits frames 3, 4, and 5.

Characteristics of Go-Back-N (GBN)

Sender Window Size (W_s)

In Go-Back-N, the sender window size (W_s) is equal to N.

For example, in GBN-10, the sender can transmit 10 frames without waiting for acknowledgements.

2. Receiver Window Size (W_R)

- Receiver window size (W_R) is always 1.
- Receiver accepts only the next expected frame.
- Frames arriving out of order are discarded.
- Receiver does not buffer out-of-order frames.

3. Acknowledgements

Acknowledgments are control messages sent by the receiver to confirm the successful reception of data frames. They play a crucial role in ensuring reliable communication. If the sender does not receive an acknowledgment within a specified timeout interval, it assumes the frame was lost or corrupted and retransmits it.

Types of ACKs

Cumulative ACK

- A single ACK confirms the receipt of all frames up to a particular sequence number.
- Used in Go-Back-N (GBN).
- Reduces acknowledgment traffic, improving efficiency.
- If an ACK is lost, multiple frames may be considered unacknowledged, causing unnecessary retransmissions.

Independent ACK

- Each frame is acknowledged separately.
- Used in Selective Repeat ARQ.
- Provides higher reliability by avoiding retransmission of correctly received frames.
- Increases control overhead due to a higher number of ACKs.

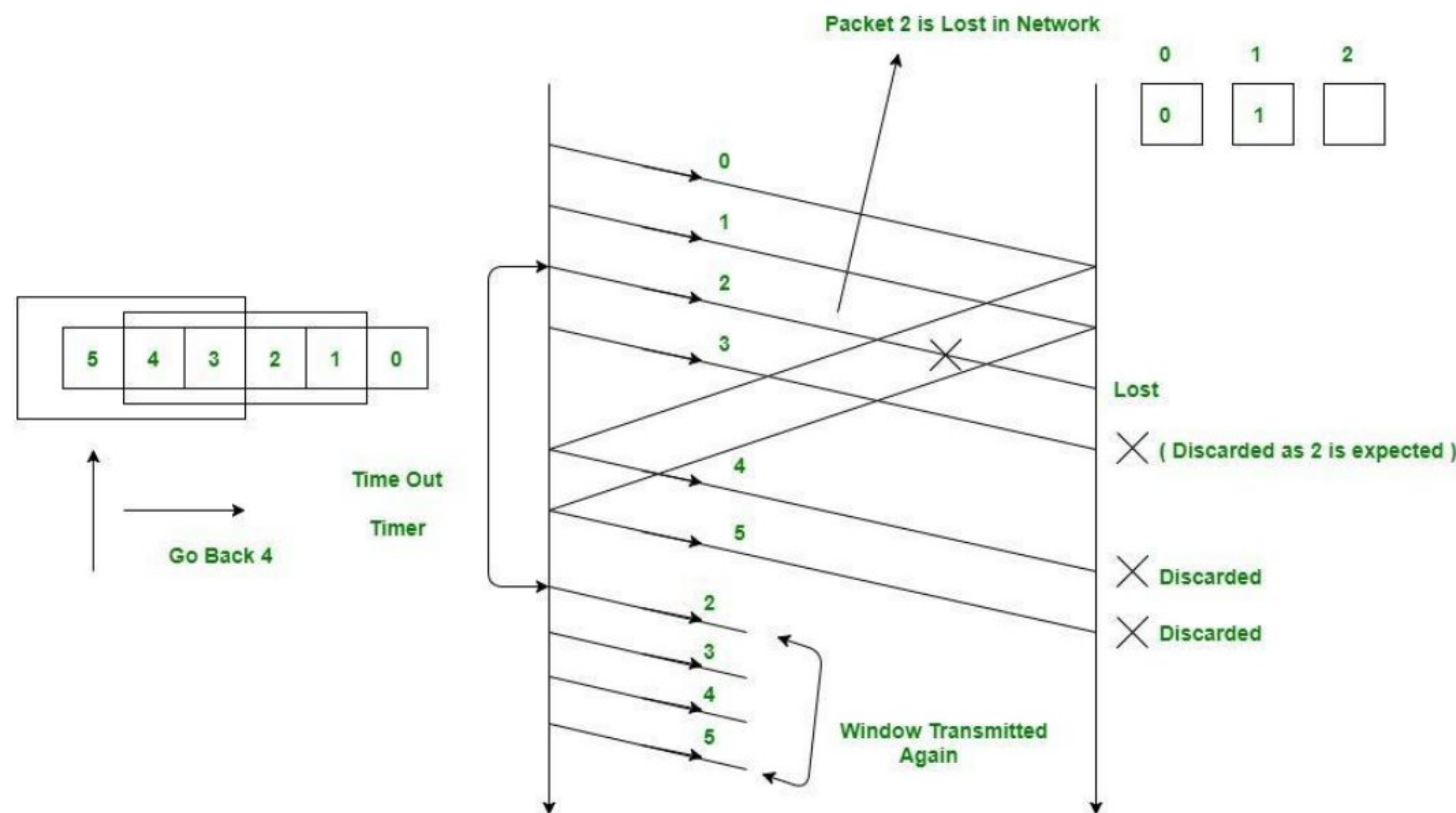
Working of GB-N Protocol

Sender Side

- The sender maintains a sliding window of size N (e.g., in GBN-4, the window size is 4).
- It can transmit up to N frames without waiting for individual ACKs.
- Each transmitted frame is associated with a timer.
- When an ACK is received, the sender slides the window forward, allowing transmission of new frames.
- If a timeout occurs for a frame, the sender retransmits that frame and all subsequent frames in the current window.

Receiver Side

- The receiver maintains a window size of 1 ($WR = 1$).
- It accepts only the next expected frame in sequence.
- If the expected frame is received correctly, then the receiver sends an ACK, and updates the expected sequence number.
- If an out-of-order frame is received, then the receiver discards the frame, and resends the ACK for the last correctly received frame.

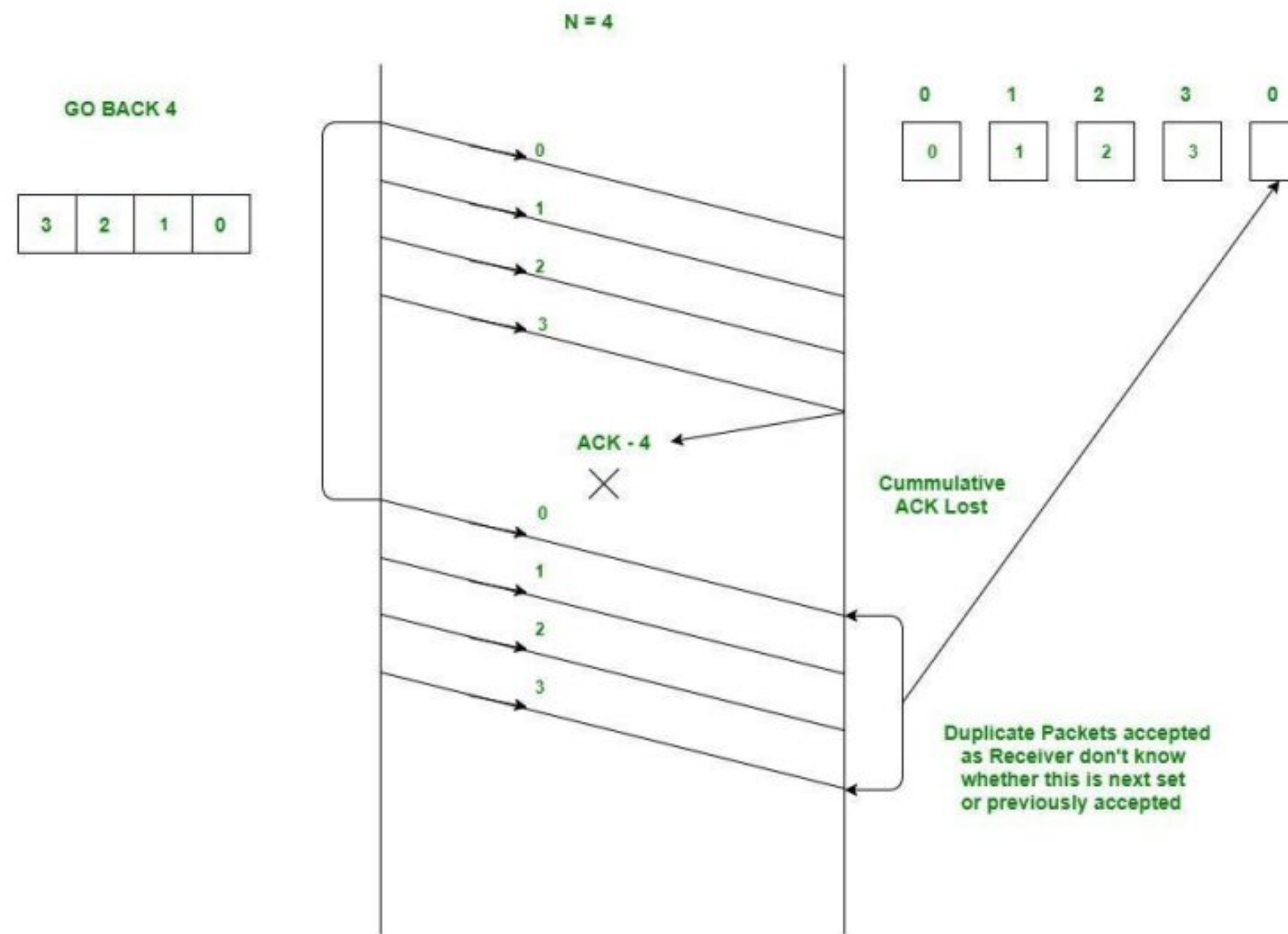


Example of GB-N Protocol

Consider an Example of GB4.

- Sender window size is 4 therefore we require a minimum of 4 sequence numbers to label each packet in the window.
- Now suppose receiver has received all the packets(0, 1, 2 and 3 sent by sender) and hence is now waiting for packet number 0 again (We can not use 4 here as we have only 4 sequence numbers available since $N = 4$).
- Now suppose the cumulative ack for the above 4 packets is lost in the network.
- On sender side, there will be timeout for packet 0 and hence all the 4 packets will be transmitted again.
- Problem now is receiver is waiting for new set of packets which should have started from 0 but now it will receive the duplicate copies of the previously accepted packets.
- In order to avoid this, we need one extra sequence number.
- Now the receiver could easily reject all the duplicate packets which were starting from 0 because now it will be waiting for packet number 4 (We have added an extra sequence number now).

This is explained with the help of the illustrations below. Trying with Sequence numbers 4.



Stop and Wait ARQ

Stop-and-Wait ARQ is a reliable data link layer protocol in which the sender transmits one frame at a time and waits for an acknowledgment (ACK) before sending the next frame. If the ACK is not received within a specified timeout period, the same frame is retransmitted to ensure reliability.

- Operates in connection-oriented communication
- Provides error control using acknowledgments and retransmissions
- Ensures flow control by allowing only one outstanding frame
- Uses two sequence numbers (0 and 1) to prevent duplication

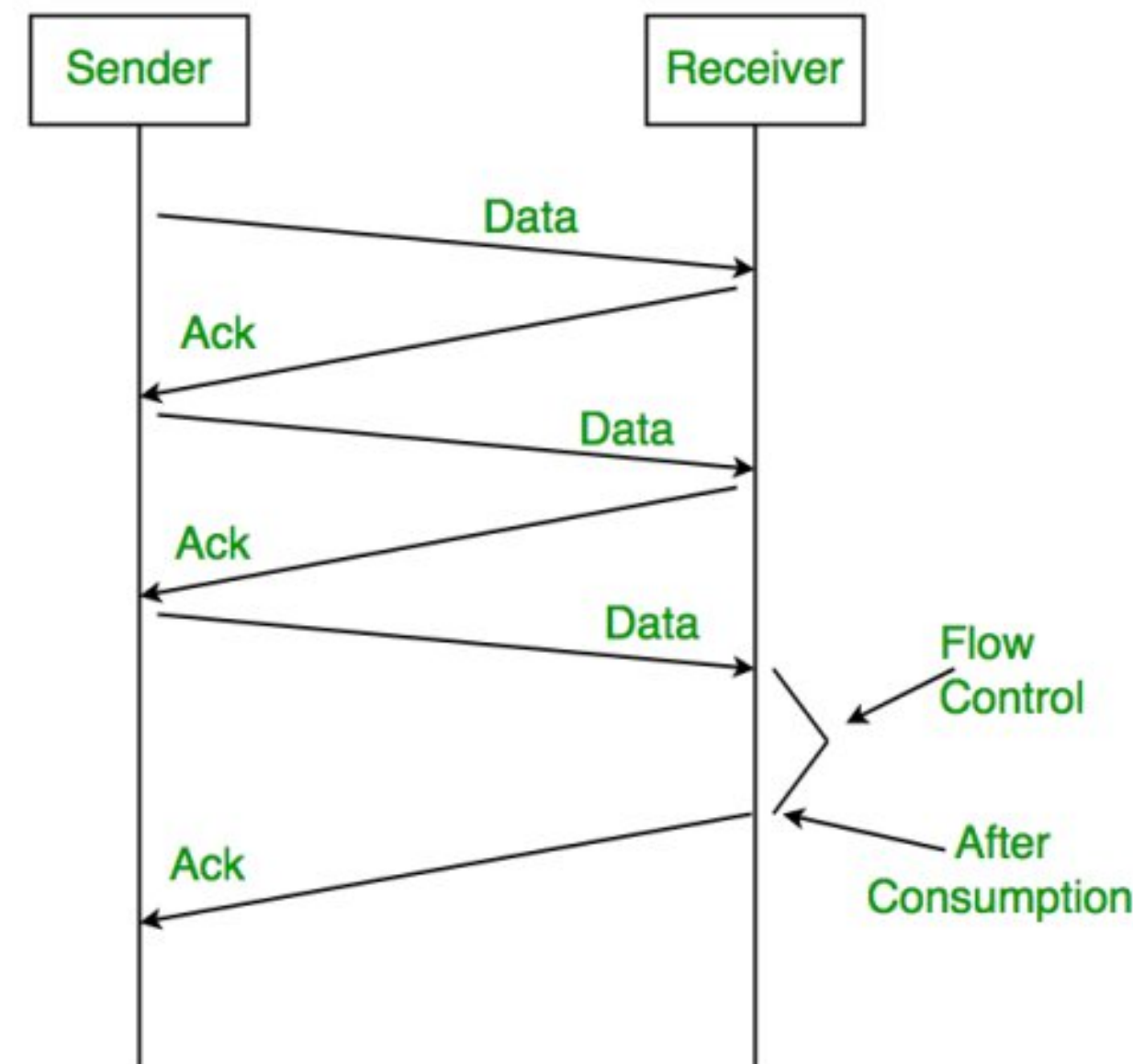
1. Simple Stop and Wait

At Sender

- Rule 1: The sender transmits only one data packet at a time.
- Rule 2: The sender sends the next packet only after receiving an acknowledgment (ACK) for the previously sent packet.

At Receiver

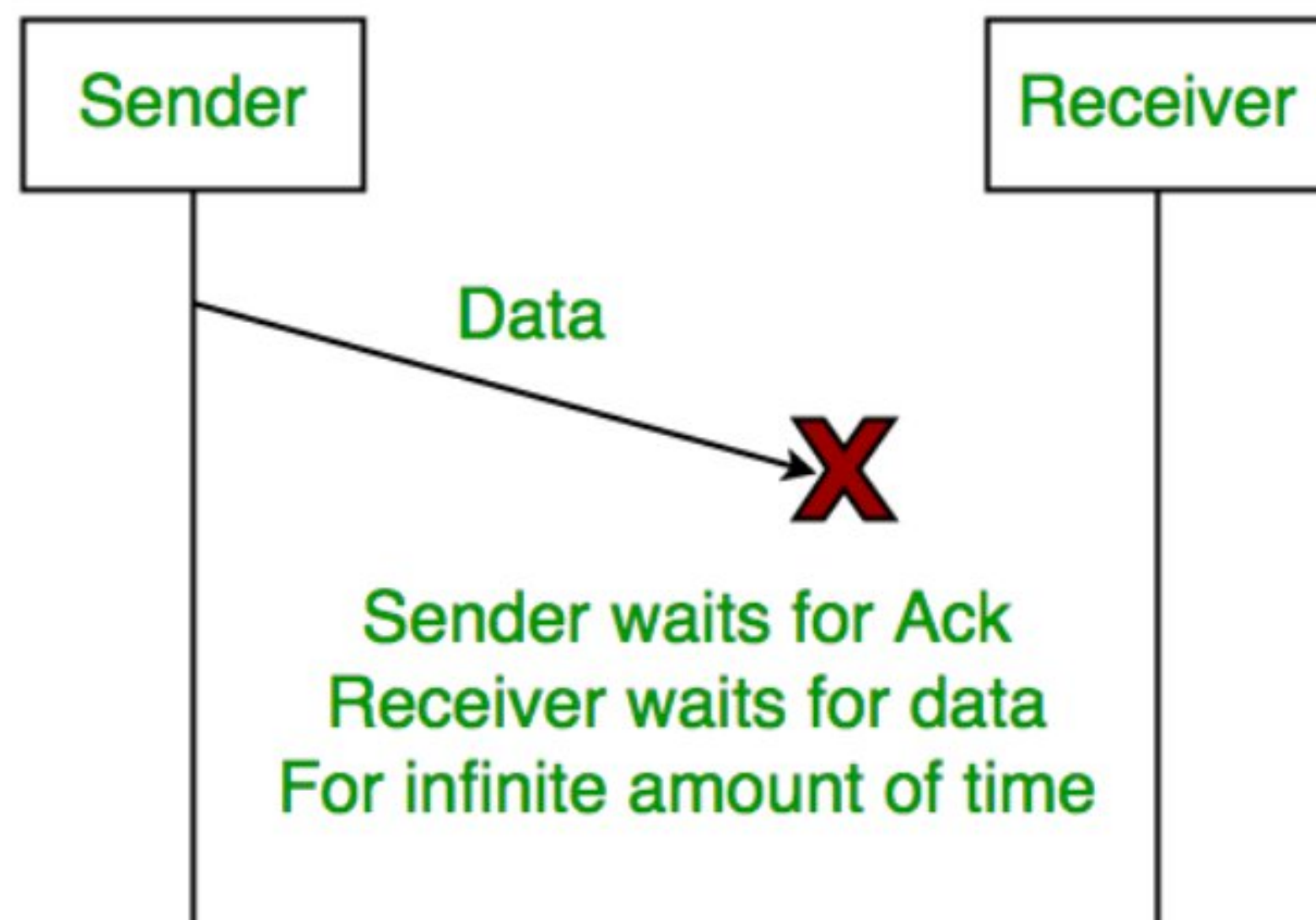
- Rule 1: The receiver sends an acknowledgment immediately after successfully receiving and processing a data packet.
- Rule 2: Acknowledgments are sent after each packet is consumed, ensuring proper flow Control flow control between sender and receiver.



Problems Associated with Stop and Wait

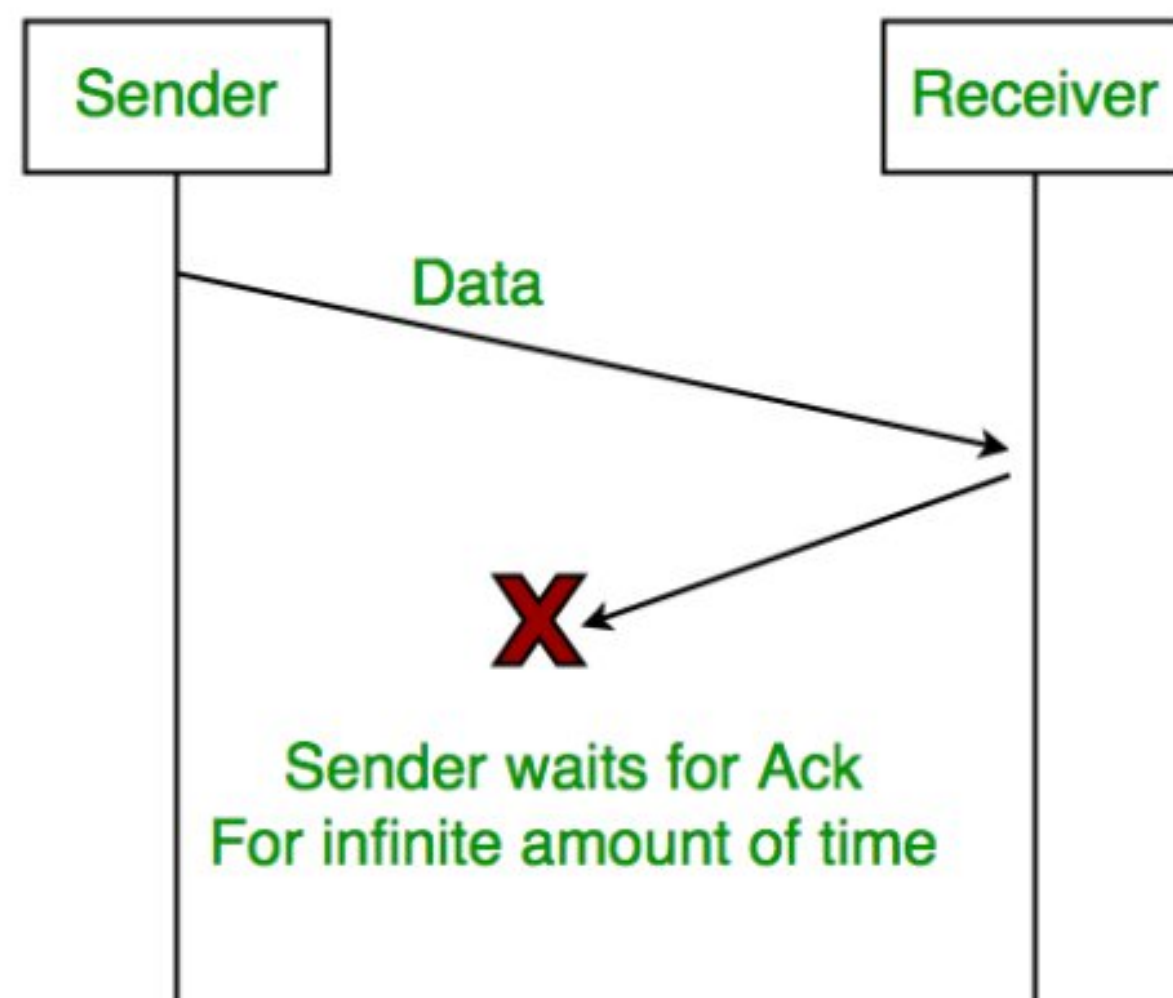
1. Lost Data

If a sender transmits a frame and it gets lost in the network, the receiver does not receive it and therefore does not send an acknowledgment (ACK). However, the sender does not wait forever. It starts a timer after transmitting the frame. If the ACK is not received before the timeout expires, the sender retransmits the same frame.



2. Lost Acknowledgement

In this case, the sender sends a data packet which is successfully received by the receiver. The receiver sends an acknowledgment, but the ACK is lost in the network. As the sender does not receive the acknowledgment, it assumes the packet was not received and does not send the next packet, because Stop-and-Wait requires an ACK before proceeding.

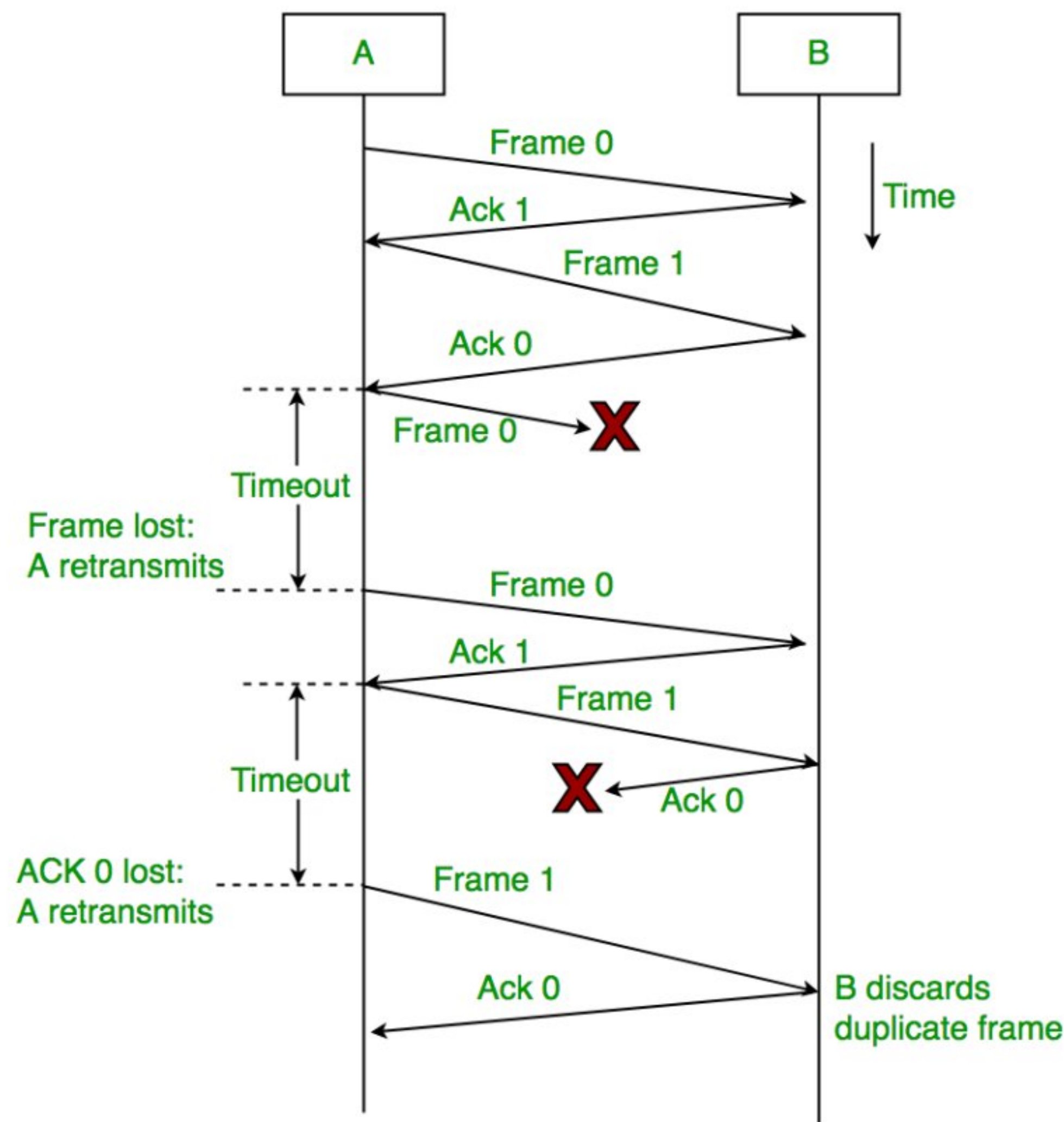


3. Delayed Acknowledgement/Data

Here, the sender sends a data packet and the receiver sends an acknowledgment, but the ACK arrives after the sender's timeout period. The sender may mistakenly interpret this delayed acknowledgment as the response to a later transmission, causing confusion and possible data duplication. This problem highlights the need for sequence numbers and timeouts in ARQ protocols.

Working of Stop and Wait for ARQ

- The sender A transmits a data frame with sequence number 0.
- After successfully receiving the frame, the receiver B sends an acknowledgment (ACK) with sequence number 1, indicating that it is expecting the next frame.
- The sender waits for this acknowledgment before sending the next data frame.
- Stop-and-Wait ARQ uses only a one-bit sequence number (0 and 1), which means only one frame can be in transit or buffered at any time at both the sender and receiver.



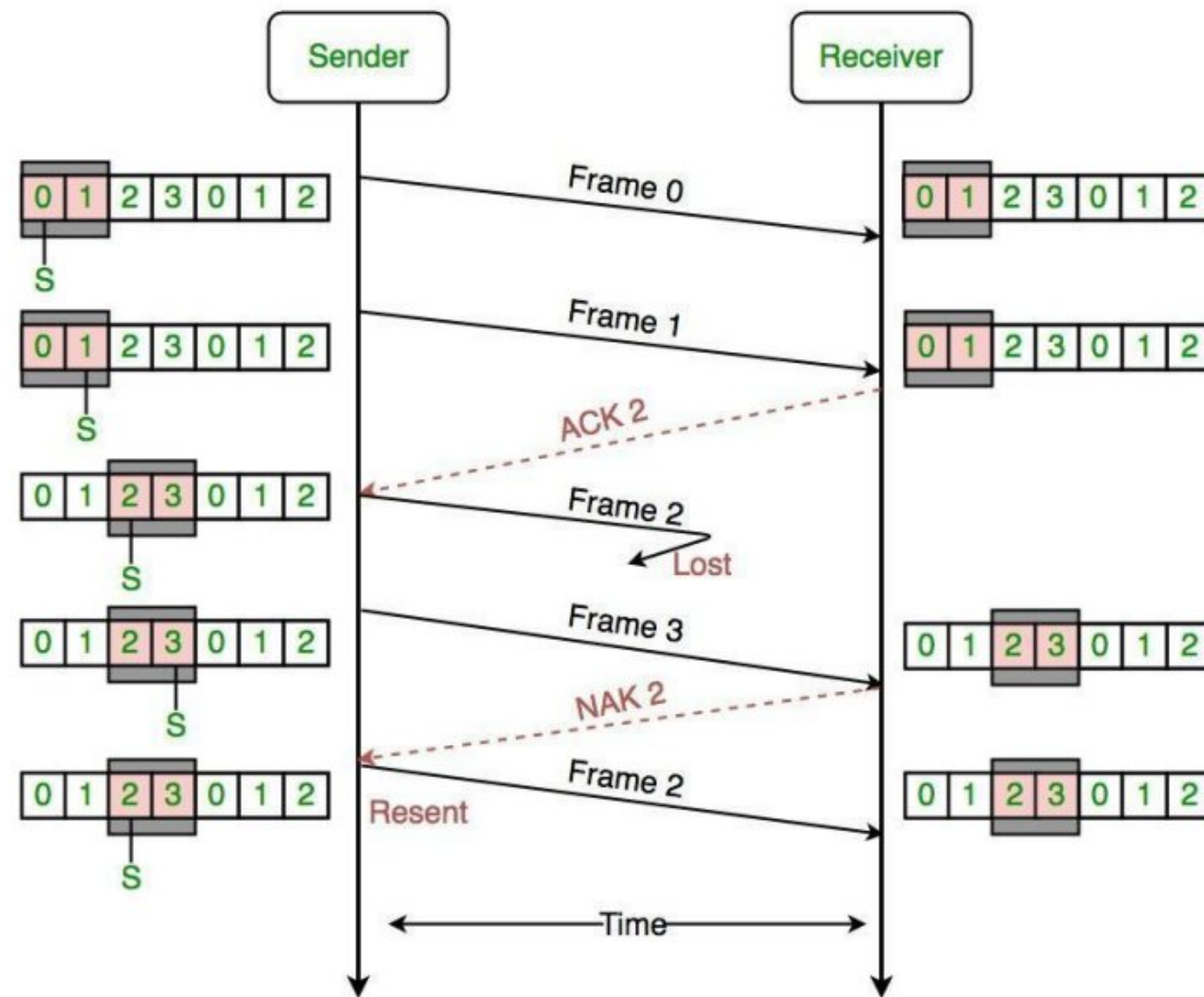
Selective Repeat - Sliding Window Protocol

The Selective Repeat Protocol (SRP) is a reliable data transmission technique that enhances protocols like Go-Back-N by retransmitting only those packets that are lost or corrupted, rather than resending all subsequent packets after an error. This selective retransmission reduces unnecessary overhead and improves efficiency, especially on unreliable or noisy communication links.

- **Window Size:** The sender's window (W_s) and the receiver's window (W_r) are of equal size. This keeps transmission and reception synchronised.
- **Packet Handling:** The receiver can accept packets that arrive out of order. Correct packets are buffered until missing ones are retransmitted.
- **Retransmission Mechanism:** The sender retransmits only the packets that are not acknowledged. Retransmission occurs after a timeout or on receiving a Negative Acknowledgement (NAK).

Why Window Size is Important

Selective Repeat Protocol (SRP): the sender and receiver window size must be less than or equal to 2^{m-1} , which is half of the available sequence number space. If the window size exceeds this limit, sequence numbers may wrap around before earlier packets are acknowledged. As a result, the receiver may be unable to distinguish between new packets and old retransmitted packets. This ambiguity can cause incorrect packet acceptance and lead to transmission errors.



Switching | Computer Networks

Switching is the process of transferring data from one device to another within the same network or across different networks using switches. It mainly occurs at the Data Link Layer (Layer 2) of the OSI model, where data is forwarded in the form of frames based on MAC addresses.

- A switch connects multiple devices within a network.
- It reduces interference by sending data only to the intended device.
- It forwards data frames to the correct port using MAC addresses.
- It delivers data directly to connected devices (computers, phones).
- For communication outside the local network, it forwards data to a router

Types of Switching

There are three types of switching methods:

Message Switching:

Message switching is one of the earliest switching techniques. In this method:

- The entire message is sent to an intermediate node.
- The message is stored temporarily at that node before being forwarded to the next node.
- No dedicated path is established between the sender and receiver.

Analogy: Like sending a physical letter that is first stored at a post office before moving to the next branch.

Circuit Switching:

Circuit switching is a connection-oriented method widely used in traditional telephone networks. In this method:

- a dedicated path is established between the sender and receiver before any data transmission begins
- The connection remains reserved for the entire duration of the communication session
- Both ends enjoy a fixed bandwidth and low latency.

Analogy: Like reserving a private road for the whole trip, even if no cars are on it for some time.

Packet Switching:

Packet switching is the foundation of modern computer networks, including the Internet. In this method:

- Data is divided into small packets.
- Each packet carries a portion of the message and the destination address.
- Packets may travel through different paths in the network.
- At the destination, packets are reassembled to reconstruct the original message.

Analogy: Like sending a book in separate chapters through different courier routes, then reassembling them at the destination.

Comparison of Switching Types

Message Switching	Circuit Switching	Packet Switching
No dedicated path; the entire message is stored and forwarded at each node.	A fixed, dedicated path is established before transmission.	Data is split into packets that follow dynamic or virtual paths.
Very inefficient due to full message storage and forwarding.	Inefficient because bandwidth remains reserved even when idle.	Highly efficient due to optimal bandwidth utilization.
High transmission delay.	Low delay after initial setup.	Delay varies based on network conditions.
Less reliable and obsolete.	Reliable with guaranteed connection.	Reliable with error control and retransmission.

Message Switching	Circuit Switching	Packet Switching
Used in telegraph systems.	Used in traditional telephone networks.	Used in the Internet.